MILITARY SPECIFICATION

# SODIUM CHLORIDE-SODIUM BICARBONATE MIXTURE, 4.5 Gm, 2s

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers a Sodium Chloride-Sodium Bicarbonate Mixture for oral use for self-treatment of burns resulting from chemical or nuclear weapons attack. (See 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-B-676 | Boxes, Set-Up. |
| PPP-C-00186a | Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. |

Military

| | |
|---|---|
| MIL-B-131 | Barrier Material, Vapor Waterproof, Flexible. |
| MIL-L-10547 | Liners, Case Waterproof. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |



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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a dry mixture formulated so that each 4.50 grams (Gm) of powder consists of 3.00 Gm of Sodium Chloride (NaCl) and 1.50 Gm of Sodium Bicarbonate ($NaHCO_3$). A tolerance in the quantity of each ingredient is permitted to the extent that the mixture (powder) shall comply with the requirements of 3.1.2. The Sodium Chloride and Sodium Bicarbonate used in the manufacture of the product shall be in accordance with the tests, standards, and requirements of the U.S.P. Shall be suitable for oral use after dissolving contents of packet in a canteen of water as an electrolyte replacement in self treatment of burns resulting from chemical or nuclear weapons attack.

3.1.1 Appearance. Shall be uniformly white and in powder form.

3.1.2 Contents. Each envelope shall assay to contain not less than 2.75 Gm and not more than 3.35 Gm of Sodium Chloride (NaCl), and not less than 1.35 and not more than 1.70 Gm of Sodium Bicarbonate ($NaHCO_3$), when assayed as specified in 4.5.1. In addition, the average weight of Sodium Chloride and of Sodium Bicarbonate, based on the assay of each lot, shall be not less than 95.0 percent and not more than 105.0 percent of the labeled amount when determined on a composite sample of 10 envelopes.

3.2 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the product to the date of delivery to the Government.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of sodium chloride-sodium bicarbonate mixture produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

| For visual examination | Inspection level | AQL (% defective) | Unit of product |
|---|---|---|---|
| Major A | II | 1.0 | Envelope |
| Major A' | S-3 | 1.0 | Envelope |
| Major B | S-3 | 1.0 | Unit package |
| Major C | II | 1.0 | Intermediate package |
| Major D | II | 1.0 | Exterior container |

For end item testing. Acceptance number shall be zero (0) for all sample sizes.

| Characteristic | Requirement paragraph | Test paragraph | Inspection level | Minimum sample |
|---|---|---|---|---|
| Contents: | | | | |
| NaCl | 3.1.2 | 4.5.1 | S-2 | 10 |
| $NaHCO_3$ | 3.1.2 | 4.5.1 | S-2 | 10 |

4.3.1 When possible, the same sample shall be used in the determination of two or more tests.

4.4 The sodium chloride-sodium bicarbonate mixture shall be examined to determine compliance with all requirements contained in this specification.



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4.4.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major A

101. Envelope not heat-sealed.
102. Excess air not expelled from envelope.
103. Label not complete and legible.

Major A'

121. Mixture not in powder form.
122. Powdered mixture not white.
123. Powdered mixture not dry (is caked or lumped).
124. Powdered mixture and immediate container not free of interaction.
125. Powdered mixture not free of contamination.

Major B

151. Material, construction, and marking of unit package not as specified.

152. Printed front side on one immediate container (envelope) and printed back side of other immediate container (envelope) not clearly visible in unit package.

Major C

161. Material, construction, and marking of exterior container not as specified.

Major D

171. Material, construction, and marking of exterior container not as specified.

*Examination is not restricted to the classified defects listed above.

4.5 Tests.

4.5.1 Assay. Select at random a quantity of filled envelopes from each lot in accordance with the sample size indicated in 4.3. Accurately determine the weight of powder in each envelope. Assay the powder obtained from the envelopes, as follows:

Dissolve the accurately weighed contents of an envelope in 50 ml of purified water. Add two drops of methyl orange and titrate with 1N sulfuric acid. Each ml of 1N sulfuric acid is equivalent to 84.01 mg of $NaHCO_3$.



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Subtract the weight of sodium bicarbonate from the weight of mixture to give the weight of sodium chloride present. A minimum of five (5) envelopes per lot shall be assayed for the above method.

As an alternate:

Dissolve the accurately weighed contents of an envelope in a 500-ml volumetric flask, dilute to mark and remove 25 ml aliquot into a glass-stoppered flask. Add 50.0 ml of 0.1N silver nitrate and slowly add (dropwise) dilute nitric acid until effervescence ceases. Add 3 ml of concentrated nitric acid, 5 ml of nitrobenzene, and 2 ml of ferric ammonium sulfate (1 Gm in 12.5 ml purified water). Shake well and titrate the excess silver nitrate with 0.1N ammonium thiocyanate. Each ml of 0.1N silver nitrate is equivalent to 5.845 of NaCl.

Subtract the weight of sodium chloride from the weight of mixture to give the weight of sodium bicarbonate present. A minimum of five (5) envelopes per lot shall be used for this method.

Any additional sampling for assaying the same lot may be composited, dissolved, and an aliquot removed comparable to that used for a single envelope. Assay procedure will proceed as in the case of a single envelope.

4.5.2 Identity. When the sodium bicarbonate is assayed, a chloride identification test (U.S.P.) shall give positive results. In the case where the sodium chloride is assayed, a bicarbonate (carbonate) identification (U.S.P.) shall give positive results.

## 5. PREPARATION FOR DELIVERY

5.1 Except as specified herein, preparation for delivery shall be in accordance with all applicable paragraphs of Interim Federal Specification PPP-C-00186a(DSA-DM), dated 15 May 1969.

5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act, and shall include the information required below:

(See page 6)



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5.2.1 Immediate containers. Each individual envelope shall have the information printed on the side designated below:

Front side of envelope:

(a) the item identification designated as "SODIUM CHLORIDE-SODIUM BICARBONATE"

(b) the quantity of active ingredients designated as "4.5 Gm"

(c) the stock number designated as "FSN 6505-663-2636" or "Stock No. 6505-663-2636"

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the date of manufacture.

Back side of envelope:

"FOR BURN CASES

Empty the contents of one envelope into canteen full of water (1 quart) - Shake well. Drink entire contents slowly over one hour period."

When the envelope is held upright on its long edge, the printing on both sides shall be upright and readable.

5.3 Packaging.

5.3.1 Unit of issue. One package, containing two envelopes, as specified, constitutes one unit of issue.

5.3.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Packaging quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 24 units | 1152 units |



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5.3.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3.3 Level A.

5.3.3.1 Immediate container. Immediate container shall be a thin envelope fabricated of waterproof flexible barrier material, meeting requirements for class 2 of MIL-B-131, except that material shall not be identified as specified in 3.6 thereof. Barrier material shall be constructed of a sheet of natural kraft paper laminated to aluminum foil with polyethylene. The inner face of the aluminum foil shall be coated with polyethylene to permit heat sealing. The dimensions of the envelope shall not exceed 2 inches in width and 3-5/8 inches in length and, in addition, shall be provided with a suitable tear-top. The layers of the laminated material shall not separate in handling, storage, and use. The envelopes shall pass the leakage test specified in 4.3.1.5 of PPP-C-00186a.

5.3.3.2 Unit package. Two (2) envelopes as specified in 5.3.3.1 shall be packaged in a polyethylene bag of the minimum dimensions necessary so that, together with one individual first aid dressing, they shall fit easily into, and can be extracted from the first aid packet case. The bag shall be fabricated of clear, layflat, polyethylene tubing having a nominal wall thickness of not less than 0.004 inch. The two envelopes shall be placed in the bag so that the printing on the front side of one envelope and the printing on the back side of the other envelope is upright and is clearly visible through the bag. Excess air shall be evacuated or expelled from the bag and closure effected by heat sealing. In addition, excess tubing shall be trimmed from the bag.

5.3.3.3 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.4 Packing.

5.4.1 Level A.

5.4.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3.

5.4.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.



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5.4.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.4.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

5.4.2 Level B.

5.4.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.5 Marking.

5.5.1 Unit package. There shall be no marking on the unit package.

5.5.2 Intermediate package. Each intermediate package shall be marked in accordance with Military Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number, and name of contractor shall be shown. Date of manufacture shall be shown in lieu of date packed. In addition, the cover of each carton shall be marked with the legend: "Insert one polyethylene bag, containing two envelopes, into the pouch, first aid, individual."

5.5.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown, and date of manufacture, in lieu of date packed.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Federal Stock No.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following Federal Stock No.:

6505-663-2636 SODIUM CHLORIDE-SODIUM BICARBONATE MIXTURE, 4.5 Gm, 2s.

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type which is normally procured to meet Military requirements.



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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-36594 FSN

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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**CAUTION**

**NOTICE TO BIDDERS/OFFERORS**

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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